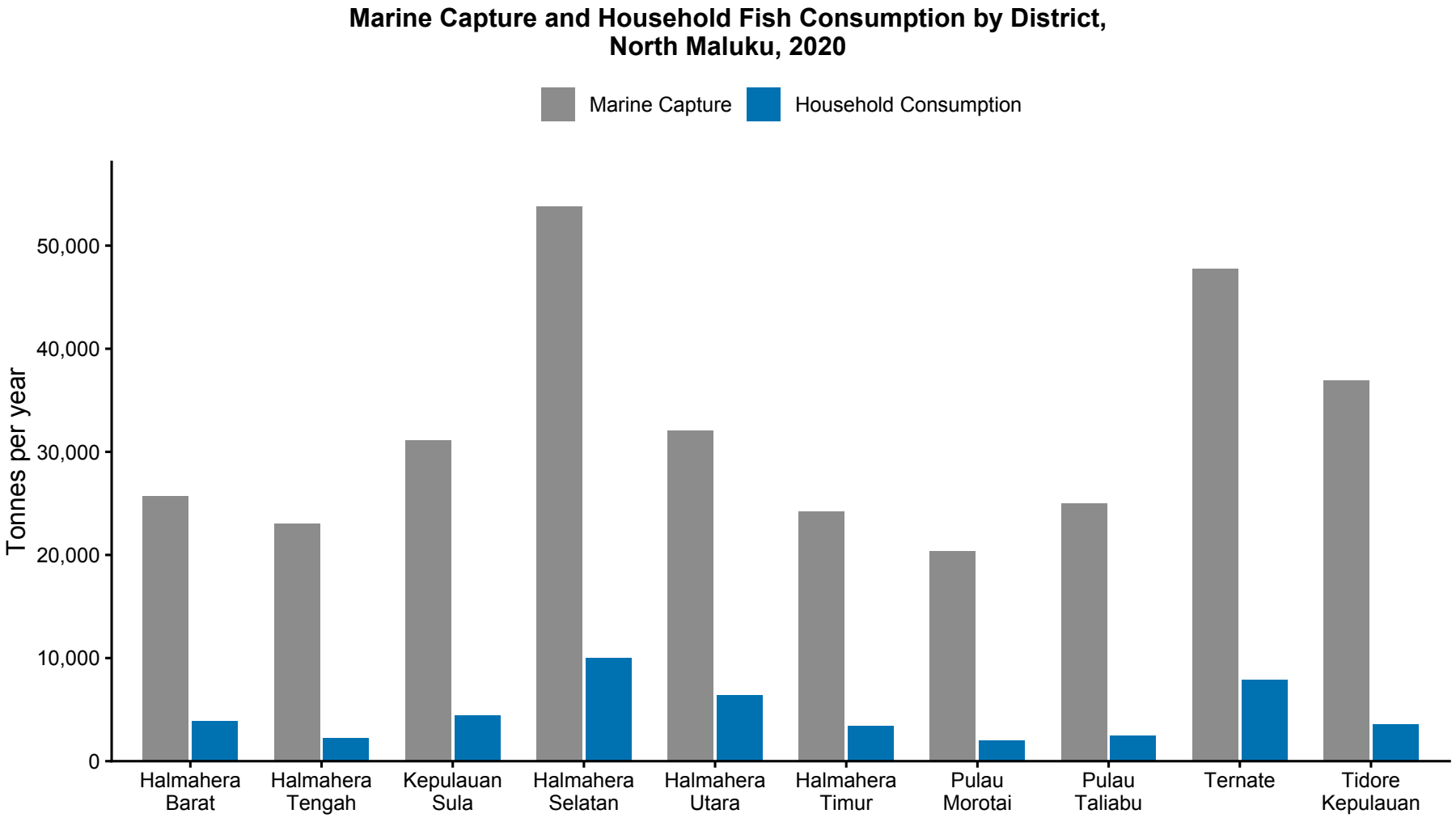


Figure 1. Marine capture exceeds household consumption in every district. Halmahera Selatan records both the largest capture (53,812 t) and the largest household consumption (10,009 t). Household consumption of the ten species groups equals 10% of the capture landed in the same district in Tidore Kepulauan and 20% in Halmahera Utara.

Chart 1 Calculation

Marine Capture and Household Fish Consumption by District, North Maluku, 2020

Formula

Capture = Σ volume_ton

Consumption = Σ (WERT × B41K9 × grams_per_unit) × 52 ÷ 10^6

Variables

Capture	=	marine capture landed in the district (tonnes per year), the grey bar
volume_ton	=	tonnes of one fish category landed in the district in 2020 (KKP 2020); the sum runs over the district's rows for the 68 categories of the ten species groups
Consumption	=	household consumption of the ten species groups in the district (tonnes per year), the blue bar
WERT	=	household weight: the number of households the sampled household represents (blok 41)
B41K9	=	quantity of the fish item the household consumed in the survey week, in the questionnaire unit (blok 41; equals B41K5 purchased + B41K7 own production)
grams_per_unit	=	grams in one questionnaire unit of the item (food_composition_units.xlsx: 1,000 for kg, 100 for ons)
Σ	=	sum over the blok 41 records of the district (R102) whose KODE is one of the 18 fish items of the ten species groups, fresh and preserved
52	=	weeks per year, scaling the March survey week to 2020
10^6	=	grams per tonne

Example

Halmahera Selatan, 2020

Capture = 7,149.14 + 6,843.84 + 4,329.30 + … + 1.08 (37 KKP categories)

= 53,811.62 t ≈ 53,812 t

Consumption = [167.6 × 3 × 1,000 + 167.6 × 2 × 1,000 + 146.5 × 2.4 × 1,000
+ … + 113.9 × 1 × 1,000] × 52 ÷ 10^6 (1,124 records)

= 192,488,814 g × 52 ÷ 10^6

= 10,009 t

Notes

- Consumption covers marine fish and squid only; non-marine aquatic foods in the survey (freshwater fish, shrimp, crab, shellfish and canned fish) amount to 841 t a year, 1.8% of household aquatic consumption, and are not included.
- Preserved fish are counted at their recorded weight, and the survey week of March 2020 is scaled to the year with the factor 52.

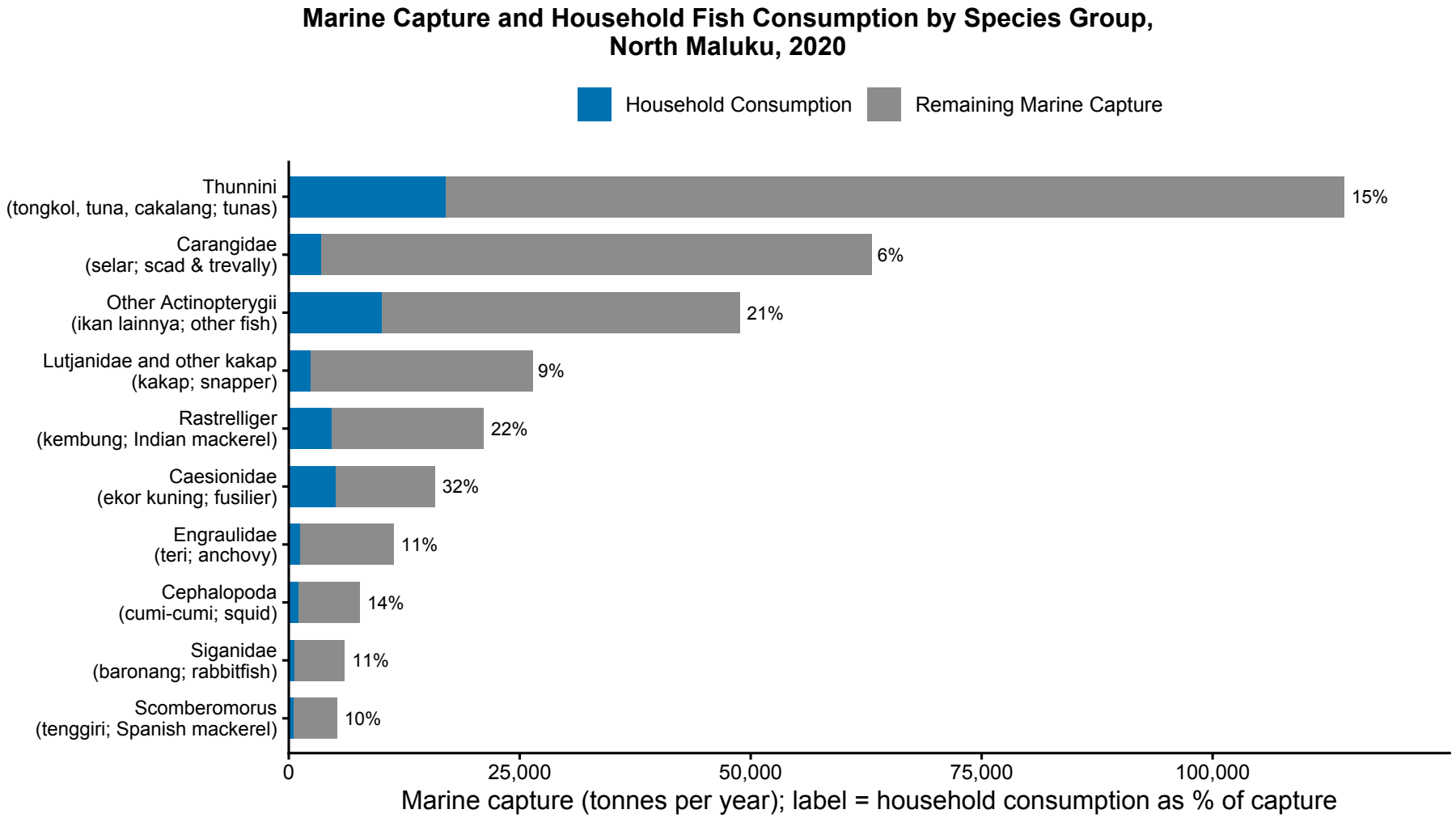


Figure 2. Thunnini (tongkol, tuna, cakalang; tunas) dominates both measures, with 114,239 t captured and 17,008 t consumed by households (15% of its capture). Across the ten groups, households consume between 6% of the capture for Carangidae (selar; scad & trevally) and 32% for Caesionidae (ekor kuning; fusilier); the remaining capture represents exports, processing, non-household uses and losses, which neither source records.

Chart 2 Calculation

Marine Capture and Household Fish Consumption by Species Group, North Maluku, 2020

Formula

Capture	=	Σ volume_ton
Consumption	=	Σ (WERT × B41K9 × grams_per_unit) × 52 ÷ 10^6
Share	=	Consumption ÷ Capture × 100

Variables

Capture	=	marine capture of the species group in all ten districts (tonnes per year), the whole bar
volume_ton	=	tonnes of one fish category landed in one district in 2020 (KKP 2020); the sum runs over the rows of the group's categories in all districts
Consumption	=	household consumption of the species group in all districts (tonnes per year), the blue part of the bar
WERT	=	household weight: the number of households the sampled household represents (blok 41)
B41K9	=	quantity of the fish item the household consumed in the survey week, in the questionnaire unit (blok 41; equals B41K5 purchased + B41K7 own production)
grams_per_unit	=	grams in one questionnaire unit of the item (food_composition_units.xlsx: 1,000 for kg, 100 for ons)
Σ	=	sum over the KKP rows of the group's categories (Capture), and over the blok 41 records whose KODE is one of the group's survey items, fresh and preserved (Consumption)
Share	=	household consumption as a percentage of marine capture, the label at the end of the bar
52	=	weeks per year, scaling the March survey week to 2020
10^6	=	grams per tonne

Example

Thunnini (tongkol, tuna, cakalang; tunas), 2020

Capture	=	24,088.44 + 11,380.37 + 6,916.88 + … + 5.52 (43 KKP rows: 7 categories in 10 districts)
	=	114,239.13 t ≈ 114,239 t
Consumption	=	[42.8 × 1 × 1,000 + 57.1 × 0.5 × 1,000 + 46.1 × 1 × 1,000 + … + 58.3 × 0.5 × 1,000] × 52 ÷ 10^6 (3,305 records)
	=	327,068,973 g × 52 ÷ 10^6
	=	17,008 t
Share	=	17,008 ÷ 114,239 × 100 = 15%

Notes

- Each KKP fish category belongs to exactly one species group through the linking file; a group's survey items are the fresh and preserved forms of its fish (Carangidae also holds item 31, bawal).
- Consumption covers marine fish and squid only; non-marine aquatic foods in the survey (freshwater fish, shrimp, crab, shellfish and canned fish) amount to 841 t a year, 1.8% of household aquatic consumption, and are not included.

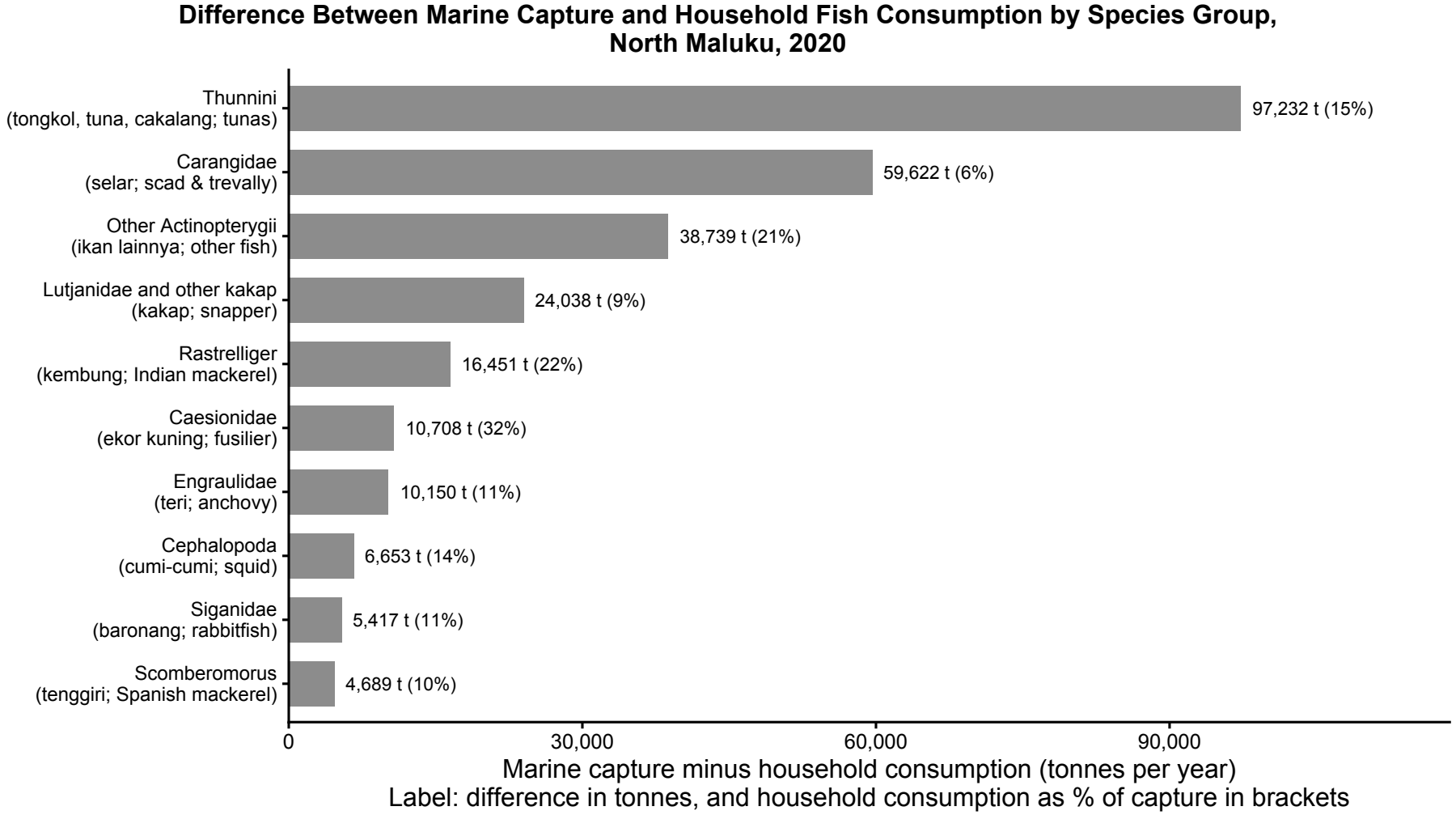


Figure 3. The absolute difference is largest for Thunnini (tongkol, tuna, cakalang; tunas) at 97,232 t, followed by Carangidae (selar; scad & trevally) at 59,622 t and Other Actinopterygii (ikan lainnya; other fish) at 38,739 t; together these three groups account for 71% of the total difference of 273,700 t. Relative to capture, the share not consumed by households ranges from 68% for Caesionidae (ekor kuning; fusilier) to 94% for Carangidae (selar; scad & trevally).

Chart 3 Calculation

Difference Between Marine Capture and Household Fish Consumption by Species Group, North Maluku, 2020

Formula

Difference	=	Capture − Consumption
Share	=	Consumption ÷ Capture × 100

Variables

Difference	=	marine capture not consumed by North Maluku households (tonnes per year), the bar
Capture	=	marine capture of the species group in all ten districts (tonnes per year), as in Chart 2
Consumption	=	household consumption of the species group in all districts (tonnes per year), as in Chart 2
Share	=	household consumption as a percentage of marine capture, the figure in brackets after the bar label

Example

Thunnini (tongkol, tuna, cakalang; tunas), 2020

Difference	=	114,239.13 − 17,007.59 = 97,231.54 t ≈ 97,232 t
Share	=	17,008 ÷ 114,239 × 100 = 15%

Notes

- The difference is fish that left the household food system of the province: exports, processing, non-household use and losses, which neither source records.
- Groups are sorted by the size of the difference; the three largest differences add to 71% of the total difference of 273,700 t.

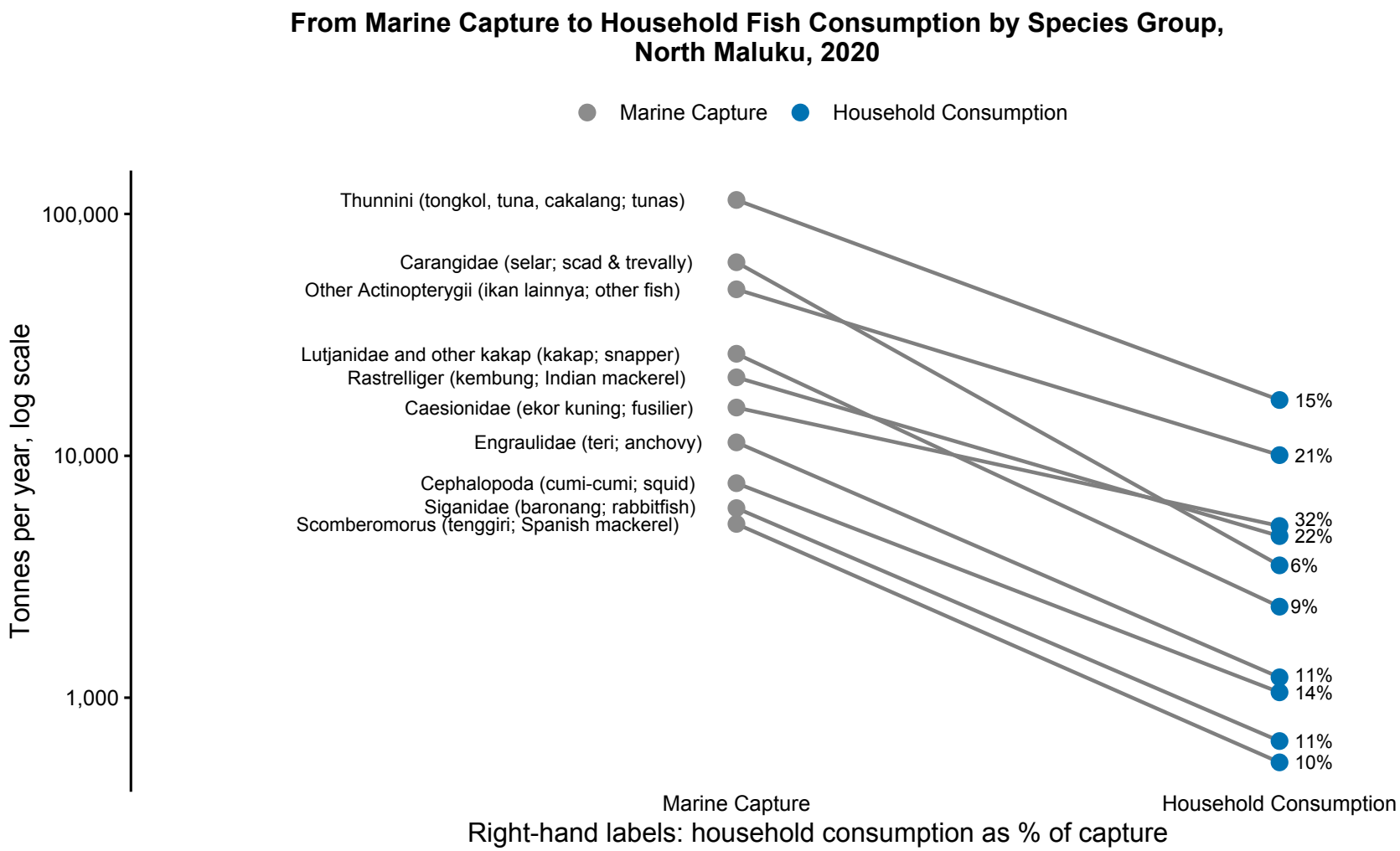


Figure 4. On a logarithmic scale the lines run nearly parallel: household consumption is a broadly similar fraction of capture in every group, between 6% and 32%. Carangidae (selar; scad & trevally) departs most from this pattern, as the second-largest capture translates into only the fifth-largest household consumption.

Chart 4 Calculation

From Marine Capture to Household Fish Consumption by Species Group, North Maluku, 2020

Formula

y_capture	=	$\log_{10}$(Capture)
y_consumption	=	$\log_{10}$(Consumption)
Drop	=	y_capture − y_consumption = $\log_{10}$(Capture ÷ Consumption)
Share	=	Consumption ÷ Capture × 100

Variables

y_capture	=	vertical position of the left point; the axis is labelled in tonnes but spaced by the logarithm
y_consumption	=	vertical position of the right point
Drop	=	vertical fall of the line; a drop of 1 means capture is ten times consumption, a drop of 2 a hundred times
Capture	=	marine capture of the species group in all ten districts (tonnes per year), as in Chart 2
Consumption	=	household consumption of the species group in all districts (tonnes per year), as in Chart 2
Share	=	household consumption as a percentage of marine capture, the label at the right end of the line
$\log_{10}$	=	logarithm to base 10

Example

Carangidae (selar; scad & trevally), 2020

y_capture	=	$\log_{10}$(63,147) = 4.80
y_consumption	=	$\log_{10}$(3,524) = 3.55
Drop	=	4.80 − 3.55 = 1.25, so Capture ÷ Consumption = $10^{1.25}$ = 17.8
Share	=	3,524 ÷ 63,147 × 100 = 5.6%

Notes

- On a logarithmic axis equal vertical distances are equal ratios, so parallel lines mean the same share of the capture is consumed.
- Lines cross where the order of the groups differs between capture and consumption (Chart 5).

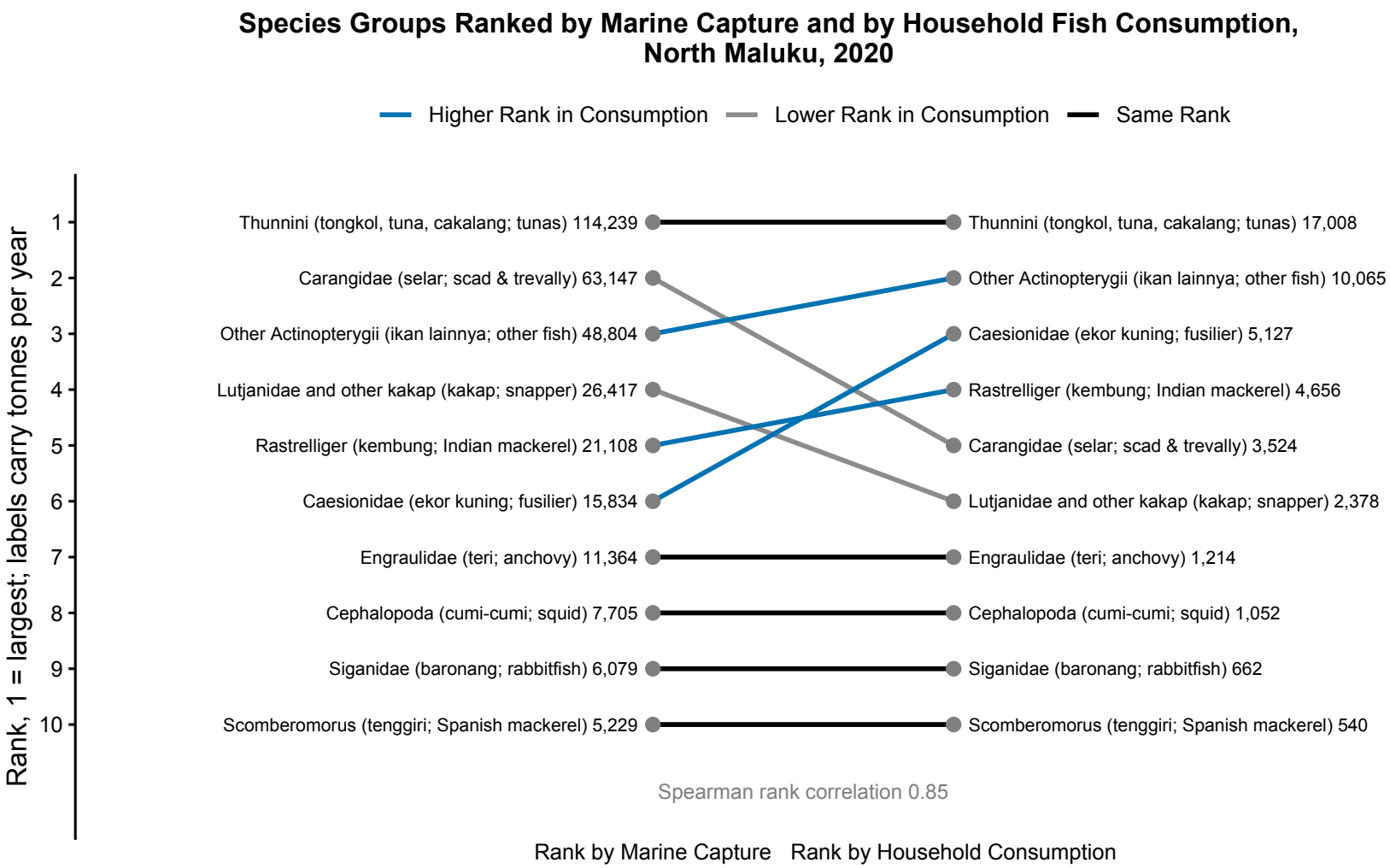


Figure 5. Rank order is largely preserved between the two measures (Spearman rank correlation 0.85). Five of the ten groups hold the same rank in both; Carangidae (selar; scad & trevally) moves from rank 2 in capture to rank 5 in consumption, Other Actinopterygii (ikan lainnya; other fish) from 3 to 2, Lutjanidae and other kakap (kakap; snapper) from 4 to 6, Rastrelliger (kembung; Indian mackerel) from 5 to 4 and Caesionidae (ekor kuning; fusilier) from 6 to 3.

Chart 5 Calculation

Species Groups Ranked by Marine Capture and by Household Fish Consumption, North Maluku, 2020

Formula

rank_capture	=	position of the group when the ten groups are sorted by Capture, 1 = largest
rank_consumption	=	position of the group when the ten groups are sorted by Consumption, 1 = largest
d	=	rank_capture − rank_consumption
ρ	=	$1 - 6 \times \Sigma d^2 \div (n \times (n^2 - 1))$

Variables

Capture	=	marine capture of the species group (tonnes per year), as in Chart 2; the left column of the chart
Consumption	=	household consumption of the species group (tonnes per year), as in Chart 2; the right column
d	=	difference in rank; 0 = the same position in both columns, a flat line
Σd^2	=	sum of the squared rank differences over the ten groups
n	=	number of species groups, 10
ρ	=	Spearman rank correlation: 1 when the two orders are identical, 0 when unrelated, −1 when reversed

Example

2020, the ten species groups (taxon names only)

d	=	Carangidae 2 − 5 = −3, Other Actinopterygii 3 − 2 = 1, Lutjanidae and other kakap 4 − 6 = −2 Rastrelliger 5 − 4 = 1, Caesionidae 6 − 3 = 3 the other five groups 0
Σd^2	=	9 + 1 + 4 + 1 + 9 = 24
ρ	=	$1 - 6 \times 24 \div (10 \times 99) = 1 - 144 \div 990 = 0.85$

Notes

- A rising blue line means the group ranks higher in consumption than in capture, a falling grey line the reverse.
- The coefficient uses positions only, not tonnes, so it says nothing about how much smaller consumption is than capture.

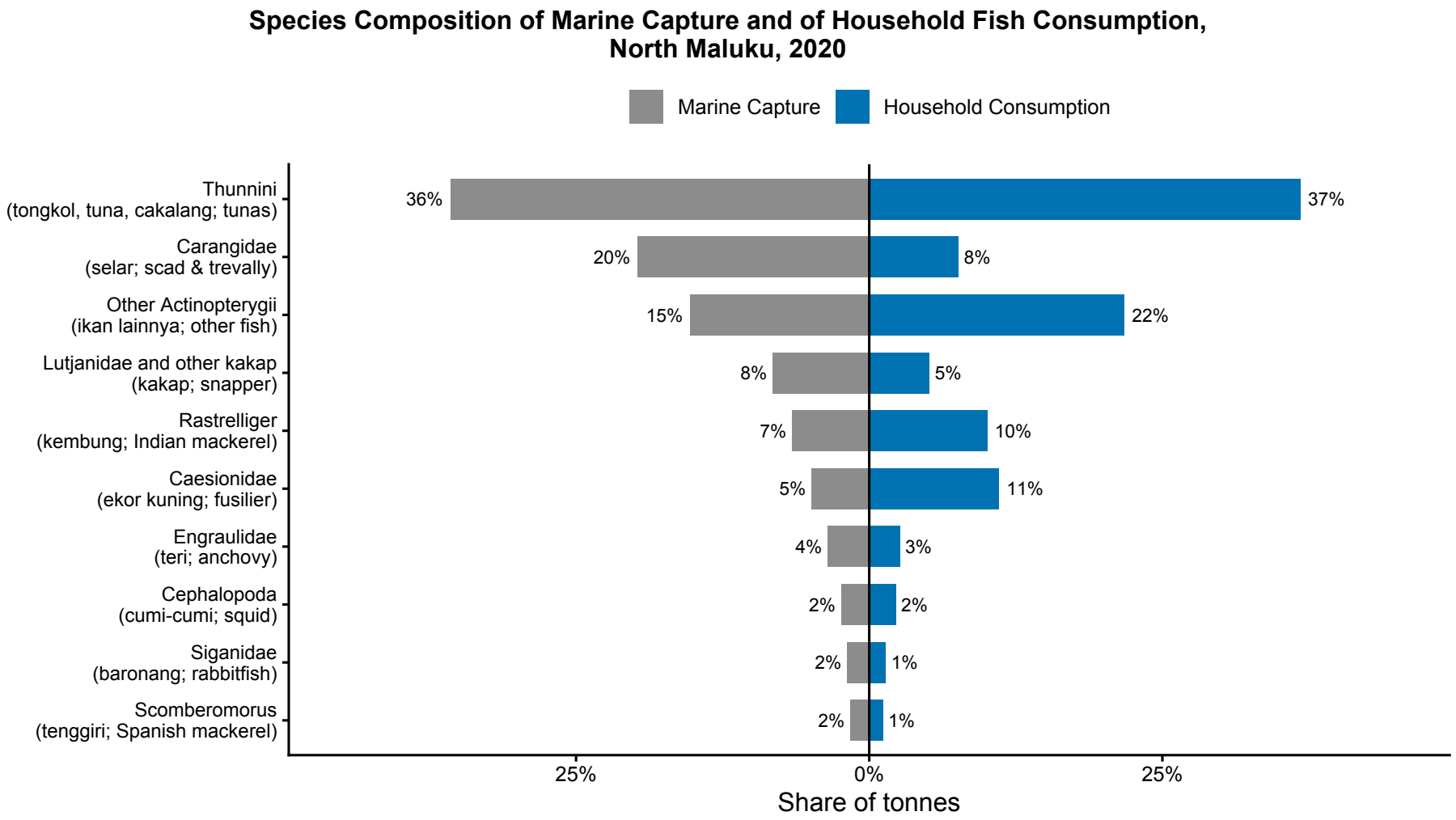


Figure 6. Thunnini (tongkol, tuna, cakalang; tunas) accounts for 36% of marine capture and 37% of household consumption, so its share is nearly the same in both. Carangidae (selar; scad & trevally) is markedly under-represented in household consumption, falling from 20% of capture to 8% of consumption, whereas Other Actinopterygii (ikan lainnya; other fish) rises from 15% to 22% and Caesionidae (ekor kuning; fusilier) from 5% to 11%.

Chart 6 Calculation

Species Composition of Marine Capture and of Household Fish Consumption, North Maluku, 2020

Formula

Share_capture = Capture ÷ Σ Capture × 100
Share_consumption = Consumption ÷ Σ Consumption × 100

Variables

Share_capture = the group's percentage of the marine capture of the ten groups, the grey bar to the left of the axis
Share_consumption = the group's percentage of the household consumption of the ten groups, the blue bar to the right
Capture = marine capture of the species group in all ten districts (tonnes per year), as in Chart 2
Consumption = household consumption of the species group in all districts (tonnes per year), as in Chart 2
Σ Capture = marine capture of the ten groups together: 319,925 t
Σ Consumption = household consumption of the ten groups together: 46,226 t

Example

Carangidae (selar; scad & trevally), 2020
Share_capture = 63,147 ÷ 319,925 × 100 = 19.7%
Share_consumption = 3,524 ÷ 46,226 × 100 = 7.6%

Notes

- The ten shares add to 100% on each side of the axis.
- A longer blue than grey bar means households consume more of the group than its share of the catch, a shorter one the reverse.

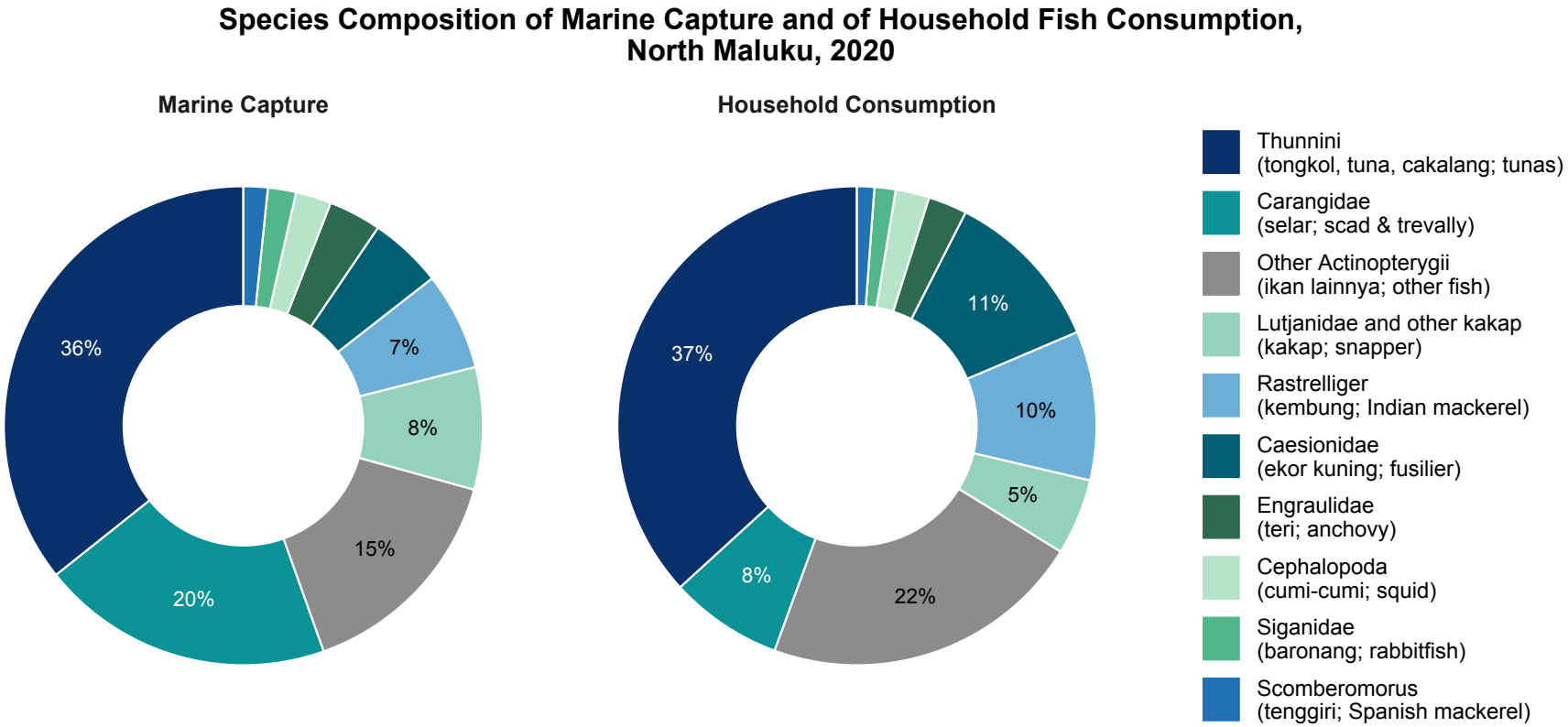


Figure 7. Three groups, Thunnini (tongkol, tuna, cakalang; tunas), Carangidae (selar; scad & trevally) and Other Actinopterygii (ikan lainnya; other fish), make up 71% of marine capture, with 36%, 20% and 15%, but 66% of household consumption, with 37%, 8% and 22%. Household consumption is spread more evenly over the remaining groups, with Other Actinopterygii (ikan lainnya; other fish) and Caesionidae (ekor kuning; fusilier) contributing 22% and 11% of the tonnes consumed.

Chart 7 Calculation

Species Composition of Marine Capture and of Household Fish Consumption, North Maluku, 2020

Formula

Share_capture	=	Capture ÷ Σ Capture × 100
Share_consumption	=	Consumption ÷ Σ Consumption × 100

Variables

Share_capture	=	the group's percentage of the marine capture of the ten groups, its segment of the grey ring
Share_consumption	=	the group's percentage of the household consumption of the ten groups, its segment of the blue ring
Capture	=	marine capture of the species group in all ten districts (tonnes per year), as in Chart 2
Consumption	=	household consumption of the species group in all districts (tonnes per year), as in Chart 2
Σ Capture	=	marine capture of the ten groups together: 319,925 t
Σ Consumption	=	household consumption of the ten groups together: 46,226 t

Example

2020, the three largest groups by capture (taxon names only)

Thunnini	=	capture 114,239 ÷ 319,925 × 100 = 35.7%; consumption 17,008 ÷ 46,226 × 100 = 36.8%
Carangidae	=	capture 63,147 ÷ 319,925 × 100 = 19.7%; consumption 3,524 ÷ 46,226 × 100 = 7.6%
Other Actinopterygii	=	capture 48,804 ÷ 319,925 × 100 = 15.3%; consumption 10,065 ÷ 46,226 × 100 = 21.8%
Together	=	70.7% of the capture ring, 66.2% of the consumption ring

Notes

- The same shares as Chart 6, in ring form; each ring adds to 100%.
- Percentages are printed for segments of at least 5%; the legend gives the colour of each group.

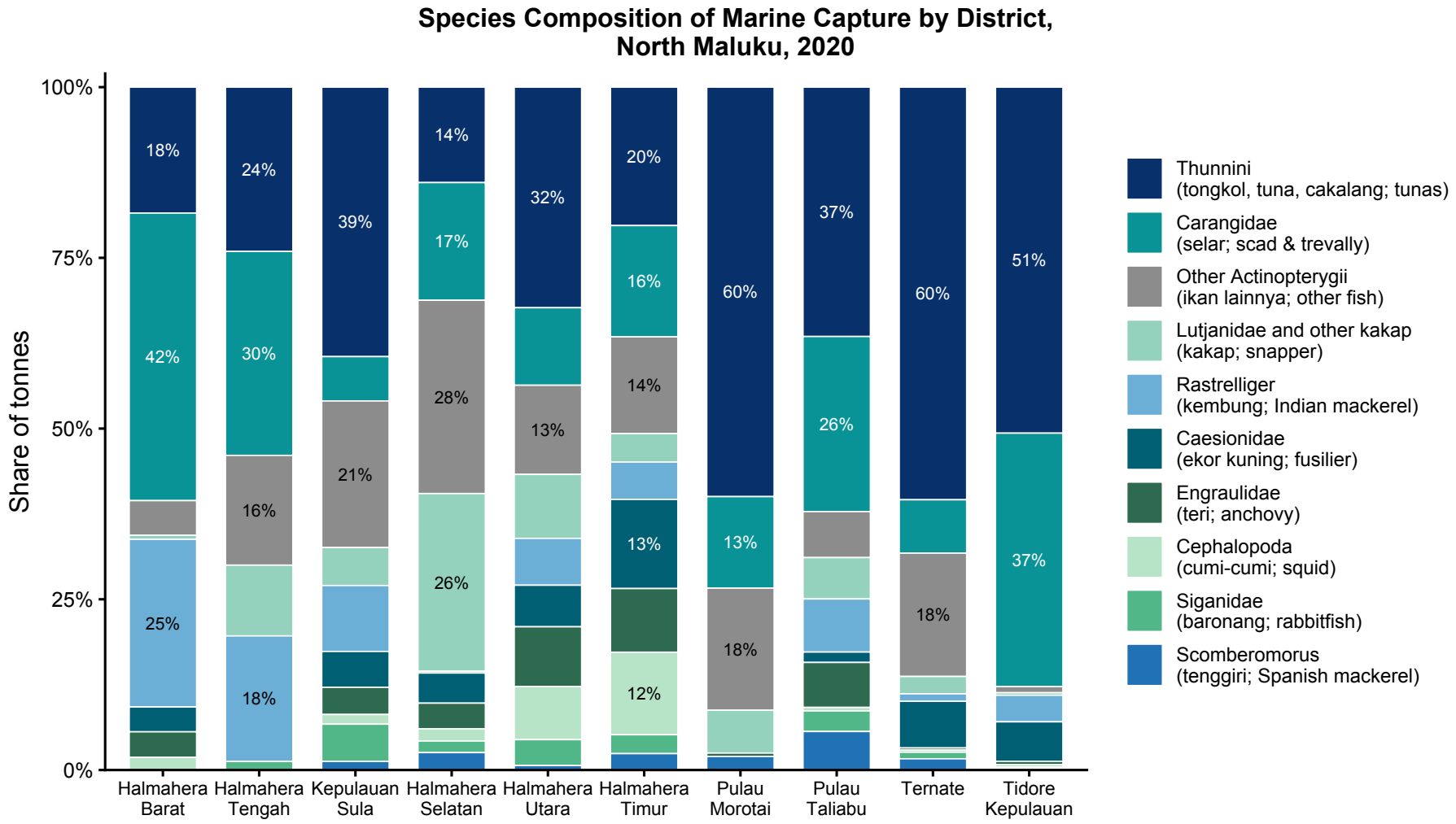


Figure 8. The species mix of marine capture varies strongly between districts. Thunnini (tongkol, tuna, cakalang; tunas) contributes 60% of capture in Pulau Morotai and Ternate but only 14% in Halmahera Selatan, where Other Actinopterygii (ikan lainnya; other fish) leads with 28%; Carangidae (selar; scad & trevally) reaches 42% in Halmahera Barat.

Chart 8 Calculation

Species Composition of Marine Capture by District, North Maluku, 2020

Formula

Capture_group	=	Σ volume_ton (the group's categories landed in the district)
Capture_district	=	Σ volume_ton (all 68 categories landed in the district), as in Chart 1
Share	=	Capture_group ÷ Capture_district × 100

Variables

Capture_group	=	marine capture of one species group landed in the district (tonnes per year)
Capture_district	=	marine capture of the ten groups landed in the district (tonnes per year), the grey bar of Chart 1
volume_ton	=	tonnes of one fish category landed in the district in 2020 (KKP 2020)
Share	=	the group's percentage of the district's capture, one segment of the stacked bar
Σ	=	sum over the KKP rows of the district that belong to the group, or to any of the ten groups

Example

Halmahera Selatan, 2020, Other Actinopterygii (ikan lainnya; other fish)

Capture_group	=	3,786.00 + 3,558.99 + 2,311.66 + … + 1.90 (11 KKP rows)
	=	15,233.34 t
Capture_district	=	53,811.62 t
Share	=	15,233 ÷ 53,812 × 100 = 28%

Notes

- The ten segments of a district add to 100%; percentages are printed for segments of at least 12%.
- District is where the fish was landed (KKP), which is not necessarily where it was consumed.

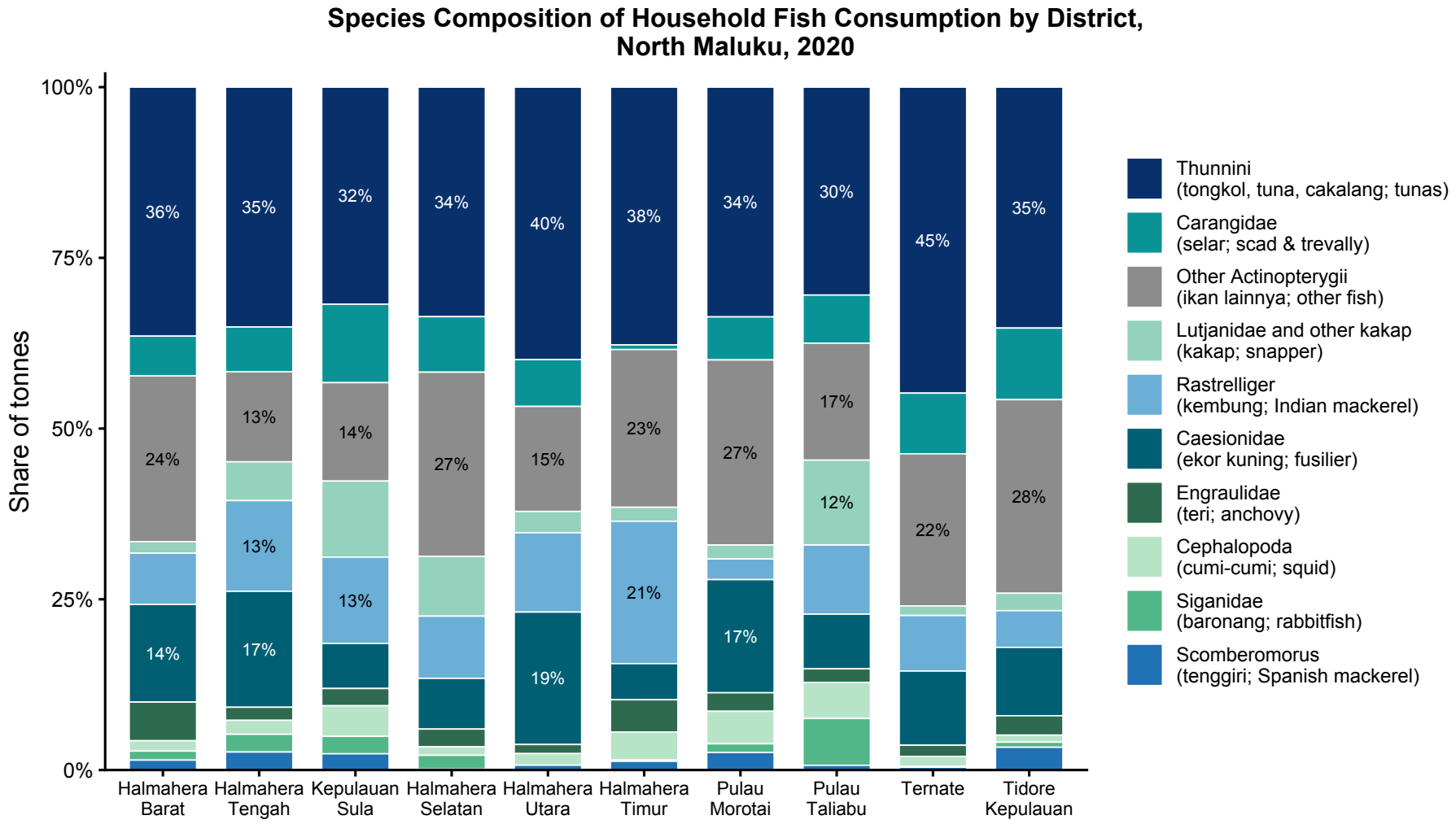


Figure 9. Household consumption is far more uniform across districts than capture: Thunnini (tongkol, tuna, cakalang; tunas) provides 30% to 45% of the tonnes consumed in every district, and Other Actinopterygii (ikan lainnya; other fish) 13% to 28%. Local preferences appear on a smaller scale, for example 21% for Rastrelliger (kembung; Indian mackerel) in Halmahera Timur and 19% for Caesionidae (ekor kuning; fusilier) in Halmahera Utara.

Chart 9 Calculation

Species Composition of Household Fish Consumption by District, North Maluku, 2020

Formula

Consumption_group = Σ (WERT × B41K9 × grams_per_unit) × 52 ÷ 10^6 (the group's items in the district)
Consumption_district = Σ (WERT × B41K9 × grams_per_unit) × 52 ÷ 10^6 (all 18 items in the district), as in Chart 1
Share = Consumption_group ÷ Consumption_district × 100

Variables

Consumption_group	=	household consumption of one species group in the district (tonnes per year)
Consumption_district	=	household consumption of the ten groups in the district (tonnes per year), the blue bar of Chart 1
WERT	=	household weight: the number of households the sampled household represents (blok 41)
B41K9	=	quantity of the fish item the household consumed in the survey week, in the questionnaire unit (blok 41; equals B41K5 purchased + B41K7 own production)
grams_per_unit	=	grams in one questionnaire unit of the item (food_composition_units.xlsx: 1,000 for kg, 100 for ons)
Share	=	the group's percentage of the district's consumption, one segment of the stacked bar
Σ	=	sum over the blok 41 records of the district (R102) whose KODE is one of the group's items, or one of all 18 items
52	=	weeks per year, scaling the March survey week to 2020
10^6	=	grams per tonne

Example

Halmahera Selatan, 2020, Thunnini (tongkol, tuna, cakalang; tunas)

Consumption_group = [167.6 × 3 × 1,000 + 146.5 × 2.4 × 1,000 + 134.9 × 2 × 1,000
+ … + 111.7 × 1 × 1,000] × 52 ÷ 10^6 (334 records)
= 64,703,096 g × 52 ÷ 10^6 = 3,365 t
Consumption_district = 10,009 t
Share = 3,365 ÷ 10,009 × 100 = 34%

Notes

- The ten segments of a district add to 100%; percentages are printed for segments of at least 12%.
- District is the district of the household (blok 41, R102), where the fish was consumed, not where it was landed.

Household Fish Consumption as a Share of Marine Capture, by District and Species Group, North Maluku, 2020

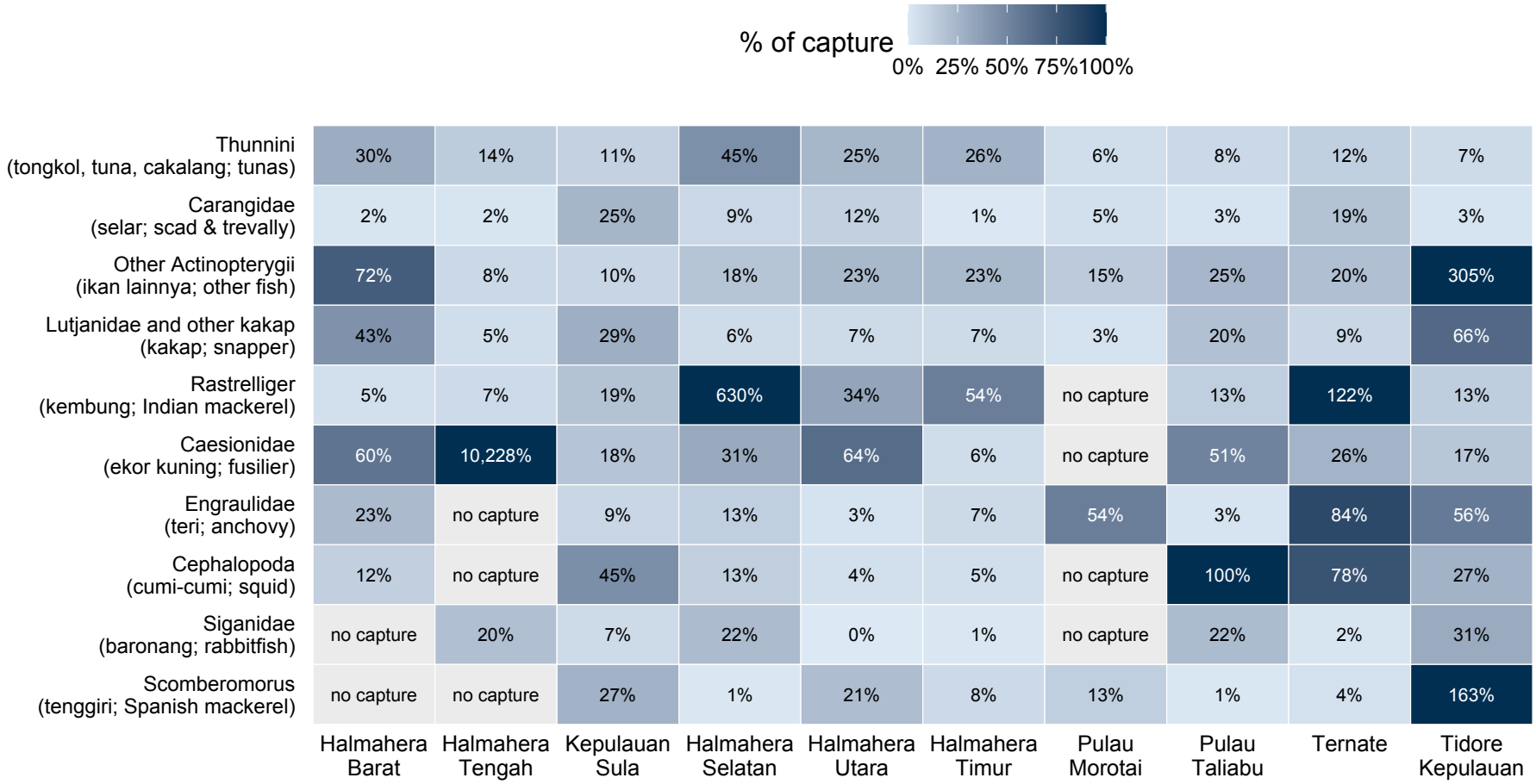


Figure 10. Within a district, household consumption of a species group is usually far below the local capture of that group. The five cells above 100%, such as Caesionidae (ekor kuning; fusilier) in Halmahera Tengah, where households consumed 382 t against a recorded capture of 4 t, mark fish eaten where KKP records little or no capture of that group; they point to trade between districts or to differences in species classification between the two sources. Carangidae (selar; scad & trevally) does not exceed 25% in any district, consistent with its low share at province level.

Chart 10 Calculation

Household Fish Consumption as a Share of Marine Capture, by District and Species Group, North Maluku, 2020

Formula

Share	=	Consumption_group ÷ Capture_group × 100
Capture_group	=	Σ volume_ton (the group's categories landed in the district), as in Chart 8
Consumption_group	=	Σ (WERT × B41K9 × grams_per_unit) × 52 ÷ 10^6 (the group's items in the district), as in Chart 9

Variables

Share	=	household consumption of the group in the district as a percentage of the marine capture of the same group landed in the same district; one cell
Capture_group	=	tonnes per year landed in the district
Consumption_group	=	tonnes per year consumed by the district's households
no capture	=	cells where KKP records no landing of the group in the district, so no share can be computed

Example

Halmahera Selatan, 2020, Thunnini (tongkol, tuna, cakalang; tunas)

Share = 3,365 ÷ 7,521 × 100 = 45%

Tidore Kepulauan, 2020, Other Actinopterygii (ikan lainnya; other fish)

Share = 1,004 ÷ 329 × 100 = 305% (more consumed than landed in the district)

Notes

- The colour scale stops at 100%, so every cell at or above 100% has the darkest shade; the printed number is the actual share.
- A share above 100% means households consumed more of the group than was landed in their district: fish brought in from other districts, or classified differently by the two sources.

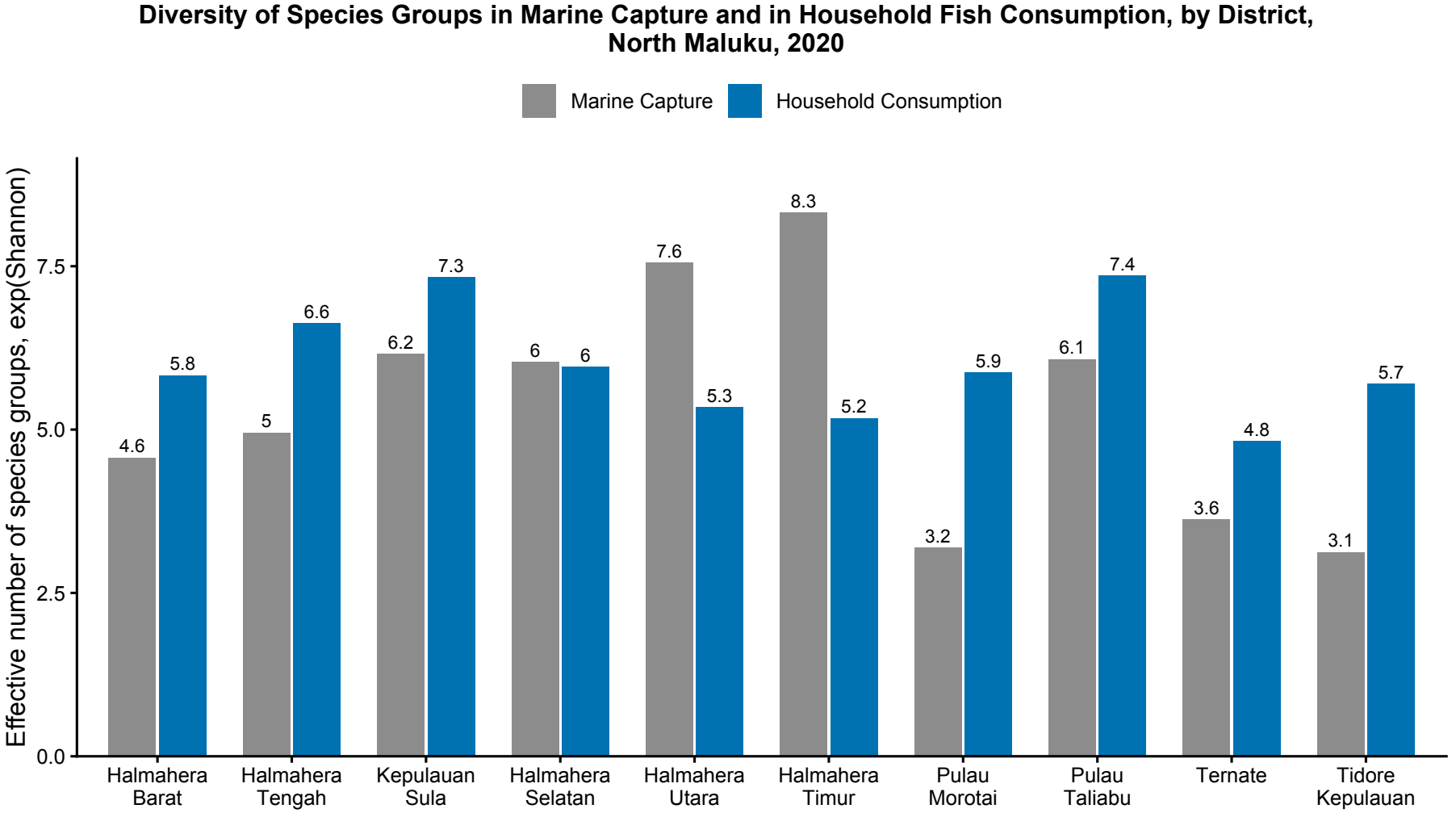


Figure 11. Household consumption is more diverse than marine capture in seven of the ten districts. The effective number of species groups ranges from 4.8 (Ternate) to 7.4 (Pulau Taliabu) in consumption and from 3.1 (Tidore Kepulauan) to 8.3 (Halmahera Timur) in capture, so diets draw on a broader set of species than the local catch alone would supply.

Chart 11 Calculation

Diversity of Species Groups in Marine Capture and in Household Fish Consumption, by District, North Maluku, 2020

Formula

p	=	Capture_group ÷ Capture_district (as a fraction, for each of the ten groups)
H	=	$- \Sigma (p \times \ln p)$ over the groups with $p > 0$
N	=	$\exp(H)$
		the same with Consumption_group and Consumption_district for the blue bar

Variables

p	=	the group's share of the district's marine capture (or household consumption), the shares of Charts 8 and 9 as fractions
H	=	Shannon entropy of the ten shares, a measure of how evenly the tonnes are spread over the groups
N	=	effective number of species groups, the bar: 1 when one group holds all the tonnes, 10 when all ten groups are equal
ln	=	natural logarithm
exp	=	the exponential function, the inverse of ln

Example

Tidore Kepulauan, 2020, marine capture

p	=	0.507, 0.371, 0.058, 0.039, 0.009, 0.005, 0.004, 0.004, 0.002, 0.002 (all ten groups have capture)
H	=	$-(0.507 \times \ln 0.507 + 0.371 \times \ln 0.371 + \ldots + 0.002 \times \ln 0.002) = 1.14$
N	=	$\exp(1.14) = 3.1$

Notes

- Groups with zero tonnes in a district do not enter H, and a group with a small share adds little, which is why N is an effective rather than a simple count.
- N is computed per district, once from the capture shares and once from the consumption shares.